JJJ-S-181b
September 3, 1964

SUPERSEDING
Int. Fed. Spec. JJJ-S-00181a (AGR-AMS)
June 11, 1962 and
Fed. Spec. JJJ-S-181
January 18, 1960

FEDERAL SPECIFICATION

SEEDS, AGRICULTURAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers agricultural seeds for use on farms, lawns, fairways, etc.

1.2 Classification.

1.2.1 *Kinds.* Agricultural seeds covered by this specification shall be of the kinds shown in table I and any desired mixtures, as specified (see 3.2.2, 6.1 and 6.3).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

PPP-B-35 — Bags: Textile, Shipping, Burlap, Cotton and Waterproof Laminated.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Service and Regulatory Announcements No. 156 of the Agricultural Marketing Serv-

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ice, United States Department of Agriculture; the Federal Seed Act and the Rules and Regulations thereunder.

State Noxious-Weed Seed Requirements Recognized in the Administration of the Federal Seed Act issued by the Agricultural Marketing Service, United States Department of Agriculture.

Regulations for Inspection and Certification of Certain Agricultural Commodities and Products Thereof, (7CFR, Part 68), issued by the Secretary of Agriculture.

(Copies of these documents may be obtained from the Seed Branch, Grain Division, Agricultural Marketing Service, United States Department of Agriculture, Washington 25, D. C.)

Applicable State seed laws. (Copies may be obtained by writing the State Department of Agriculture in the State concerned.)

3. REQUIREMENTS

3.1 Federal Seed Act and State seed laws. Agricultural seeds covered by this specification shall conform to the requirements of the Federal Seed Act and to the State seed laws as applicable. The pertinent requirements under the Federal Seed Act for labeling seed in interstate commerce require the lot number, name of the kind or kind and variety, percentages of pure seed, other crop seeds, weed seeds, inert matter, percentage of germination, and hard seeds (if any), the date tested, names and rates of occurrence of noxious-weed seeds in accordance with the law of the State into which shipped, the name and address of the shipper, and, if treated, a statement to that effect, the chemical or commonly accepted coined name of the substance with which treated, and an appropriate caution statement.

3.2 Quality requirements. Agricultural seeds shall conform to the quality requirements shown in table I.

TABLE I.—*Quality requirements for agricultural seeds*

Name of seed	Minimum percent pure seed	Minimum percent germination and hard seed	Maximum percent weed seed
Alfalfa—Medicago sativa	99	90*	.50
Alfilaria—Erodium cicutarium	95	70	1.00
Alyceclover—Alysicarpus vaginalis	98	90*	1.00
Bahiagrass—Paspalum notatum,			
Argentine	80	70	1.00
Pensacola	80	70	1.00
Barley—Hordeum vulgare	99	90	.10
Bean:			
Adzuki—Phaseolus angularis	99	90*	.00
Field—Phaseolus vulgaris	99	85*	.00
Mung—Phaseolus aureus	99	90*	.00
Beet, field and sugar—Beta vulgaris	98	80	.00
Beggarweed—Dasmodium tortuosum	95	80*	.50
Bentgrass:			
Colonial (including Astoria and Highland)—			
Agrostis tenuis	98	85	.50
Creeping—Agrostis palustris	98	85	.50
Velvet—Agrostis canina	98	85	.50
Bermudagrass:			
Common—Cynodon dactylon	97	85	1.00
Giant—Cynodon sp.	97	85	1.00
Bluegrass:			
Bulbous—Poa bulbosa	96	80	.50
Canada—Poa compressa	80	80	.50
Glaucantha—Poa glaucantha	80	80	.50
Kentucky—Poa pratensis	85	80	.50
Nevada—Poa nevadensis	85	80	.50
Rough—Poa trivialis	85	80	.50
Texas—Poa arachnifera	85	80	.50
Wood—Poa nemoralis	80	80	1.00

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TABLE I.—*Quality requirements for agricultural seeds (cont'd.)*

Name of seed	Minimum percent pure seed	Minimum percent germination and hard seed	Maximum percent weed seed
Bluestem:			
Big—Andropogon gerardi	50	50	1.00
Little—Andropogon scoparius	50	50	1.00
Sand—Andropogon hallii	50	50	1.00
Yellow—Andropogon ischaemum	50	50	1.00
Brome:			
Field—Bromus arvensis	95	80	1.00
Mountain—Bromus marginatus	92	85	1.00
Smooth—Bromus inermis	90	85	.50
Broomcorn—Sorghum vulgare technicum	98	85	.10
Buckwheat—Fagopyrum esculentum	97	85	1.00
Buffalograss—Buchloe dactyloides	75	70	1.00
Buffelgrass—Pennisetum ciliare	70	70	.50
Burclover, spotted—Medicago arabica			
(in bur)	80	90*	.50
(out of bur)	98	90*	.50
Burclover, California—Medicago hispida			
(in bur)	90	90*	.50
(out of bur)	98	90*	.50
Burnet—Sanguisorba minor	98	70	1.00
Buttonclover—Medicago orbicularis	97	85*	1.00
Canarygrass—Phalaris canariensis	98	80	.50
Canarygrass, Reed—Phalaris arundinacea	97	70	.50
Carpetgrass—Axonopus affinis	92	85	.50
Castorbean—Ricinus communis	99	80	.00
Chess, soft—Bromus mollis	92	85	1.00
Chickpea—Cicer arientinum	99	90	.00
Clover:			
Alsike—*Trifolium* hybridum	98	90*	.50
Berseem—Trifolium alexandrinum	98	90*	.50
Cluster—Trifolium glomeratum	95	85*	.50
Crimson—Trifolium incarnatum	98	85*	.50
Kenya—*Trifolium semipilosum*	98	85*	.50
Ladino—Trifolium repens	98	90*	.50
Large hop—Trifolium procumbens	98	85*	.50
Lappa—Trifolium lappaceum	98	75*	.50
Persian—*Trifolium resupinatum*	95	85*	.50
Red—Trifolium pratense	98	90*	.50
Rose—Trifolium hirtum	98	90*	.50
Strawberry—Trifolium fragiferum	97	90*	.50
Subterranean—*Trifolium subterraneum*	98	90*	.50
Small hop (suckling)—Trifolium dubium	92	85*	.50
White—Trifolium repens	98	90*	.50
Corn:			
Field—Zea mays	99	90	.00
Pop—Zea mays everta	99	90	.00
Cotton—Gossypium hirsutum	99	80	.00
Cowpea—Vigna sinensis	98	85*	.00
Crested dogtail—Cynosurus cristatus	98	80	.50
Crotalaria:			
Lance—Crotalaria lanceolata	99	80*	.50
Showy—Crotalaria spectabilis	99	80*	.50
Slender—Crotalaria intermedia	99	80*	.50
Striped—Crotalaria mucronata	99	80*	.50

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TABLE I.—*Quality requirements for agricultural seeds (cont'd.)*

Name of seed	Minimum percent pure seed	Minimum percent germination and hard seed	Maximum percent weed seed
Crotalaria: (cont'd.)			
Sunn—Crotalaria juncea	98	80*	.50
Crownvetch—Coronilla varia	98	85*	.50
Dallisgrass—Paspalum dilatatum	70	70	1.00
Dichondra—Dichondra repens	99	85*	.25
Dropseed, sand—Sporobolus cryptandrus	95	75	1.00
Emmer—Triticum dicoccum	99	90	.25
Fescue:			
Chewings—Festuca rubra var.	98	85	.50
Hair—Festuca capillata	96	85	1.00
Hard—Festuca ovina	96	85	1.00
Meadow—Festuca elatior	98	90	1.00
Red—Festuca rubra var.	98	85	.50
Sheep—Festuca ovina	96	85	.50
Tall—Festuca elatior var. arundinacea	98	90	1.00
Flax—Linum usitatissimum	97	85	.50
Grama:			
Blue—Bouteloua gracilis	66	60	1.00
Side-oats—Bouteloua curtipendula	50	50	1.00
Guar—Cyamopsis tetragonoloba	98	70*	.50
Guineagrass—Panicum maximum	75	30	1.00
Hardinggrass—Phalaris tuberosa	90	60	.50
Hemp—Cannabis sativa	98	85	.25
Indiangrass, yellow—Sorghastrum nutans	70	70	1.00
Indigo, hairy—Indigofera hirsuta	98	85*	.50
Japanese lawngrass—Zoysia japonica	70	50	1.00
Johnsongrass—Sorghum halepense	98	70	.50
Kudza—Pueraria thunbergiana	99	70*	.50
Lentil—Lens culinaris	98	80*	.50
Lespedeza:			
Korean—Lespedeza stipulacea	97	85*	1.00
Sericea or Chinese—Lespedeza cuneata	98	85*	.50
Siberian—Lespedeza hedysaroides	97	90*	.50
Striate:			
Common—Lespedeza stipulacea	96	85*	1.00
Kobe—Lespedeza stipulacea	97	85*	1.00
Lovegrass:			
Sand—Eragrostis trichodes	92	70	1.00
Weeping—Eragrostis curvula	95	80	1.00
Lupine:			
Blue—Lupinus angustifolius	99	90*	.00
White—Lupinus albus	99	90*	.00
Yellow—Lupinus luteus	99	90*	.00
Manilagrass—Zoysia matrella	97	50	.50
Meadow, foxtail—Alopecurus pratensis	90	80	.50
Medick, black—Medicago lupulina	98	90*	.50
Millet:			
Browntop—Panicum ramosum	98	85	.50
Foxtail—Setaria italica	98	85	.50
Japanese—Echinochloa crusgalli frumentacea	98	85	.50
Pearl—Pennisetum glaucum	98	85	.50
Proso—Panicum miliaceum	98	85	.50
Molassesgrass—Melinis minutiflora	50	50	1.00
Mustard:—Brassica juncea	98	85	.50
Black—Brassica nigra	98	85	.50

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TABLE I.—*Quality requirements for agricultural seeds (cont'd.)*

Name of seed	Minimum percent pure seed	Minimum percent germination and hard seed	Maximum percent weed seed
Mustard: (cont'd.)			
White—Brassica hirta	98	85	.50
Napiergrass—Pennisetum purpureum	90	70	1.00
Oat—Avena sativa and A. byzantina	98	90	.10
Oatgrass, tall—Arrhenatherum elatius	85	80	1.00
Orchardgrass—Dactylis glomerata	90	90	1.00
Panicgrass:			
Blue—Panicum antidotale	80	60	.50
Green—Panicum maximum var. trichoglume	97	80	.50
Peanut—Arachis hypogaea	99	80*	.00
Pea, field—Pisum sativum arvense	99	90*	.00
Rape:			
Annual—Brassica napus var. annua	98	85	.50
Bird—Brassica campestris	98	85	.50
Turnip—Brassica campestris	98	85	.50
Winter—Brassica napus	99	90	.50
Redtop—Agrostis alba	92	85	1.00
Rescuegrass—Bromus catharticus	97	85	1.00
Rhodesgrass—Chloris gayana	60	60	1.00
Rice—Oryza sativa	99	90	.50
Ricegrass, Indian—Oryzopsis hymenoides	95	70	.50
Roughpea—Lathyrus hirsutus	98	90*	.00
Rye—Secale cereale	97	85	.10
Ryegrass:			
Italian or Annual—Lolium multiflorum	98	90	.50
Perennial—Lolium perenne	98	90	.50
Wimmera—Lolium rigidum	98	90	.50
Safflower—Carthamus tinctorius	98	80	.50
Sainfoin—Onobrychis viciaefolia	98	75*	.50
Sesame—Sesamum indicum	98	80	.50
Sesbania—Sesbania exaltata	99	90*	.25
Smilo—Oryzopsis miliacea	90	70	.50
Sorghum:			
Grain—Sorghum vulgare	98	85	.25
Sweet (Sorgo)—Sorghum vulgare	98	85	.25
Sorghum almum—Sorghum almum	98	70	.50
Sorghum—sudangrass hybrid—Sorghum vulgare X S. sudanense	98	85	.25
Sorgrass—Sorghum sp.	98	80	.50
Sourclover—Melilotus indica	98	90*	.50
Soybean—Glycine max	98	85*	.00
Spelt—Triticum spelta	99	90	.25
Sudangrass—Sorghum vulgare sudanense	98	85	.25
Sunflower (Cult)—Helianthus annuus	99	90	.00
Sweetclover:			
White blossom—Melilotus alba	98	88*	.50
Yellow blossom—Melilotus officinalis	98	88*	.50
Sweet vernalgrass—Anthoxanthum odoratum	90	80	.50
Switchgrass—Panicum virgatum	70	50	.50
Timothy—Phleum pratense	99	85	.25
Tobacco—Nicotiana tobacum	98	75	.25
Trefoil:			
Big—Lotus uliginosus	97	85*	.50
Birdsfoot—Lotus corniculatus	97	85*	.50
[illegible]—Paspalum [illegible]ei	70	60	1.00

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TABLE I.—*Quality requirements for agricultural seeds (cont'd.)*

Name of seed	Minimum percent pure seed	Minimum percent germination and hard seed	Maximum percent weed seed
Veldtgrass—Ehrharta calycina	60	60	1.00
Velvetbean—Stizolobium deeringianum	99	80*	.00
Velvetgrass—Holcus lanatus	95	85	.50
Vetch:			
Common—Vicia sativa	98	90*	.25
Hairy—Vicia villosa	98	90*	.25
Hungarian—Vicia panonica	98	90*	.25
Monantha—Vicia articulata	98	90*	.25
Narrowleaf—Vicia angustifolia	98	90*	.25
Purple—Vicia atropurpurea	98	90*	.25
Woollypod—Vicia dasycarpa	98	90*	.25
Wheat, common, club, durum, Polish, poulard—Triticum spp.	99	90	.10
Wheatgrass:			
Beardless—Agropyron inerme	95	85	.50
Crested or fairway crested—Agropyron cristatum	95	85	.25
Crested or standard crested—Agropyron desertorum	95	85	.25
Intermediate—Agropyron intermedium	95	85	.50
Pubescent—Agropyron tricophorum	95	85	.50
Siberian—Agropyron sibiricum	95	85	.50
Slender—Agropyron trachycaulum	95	85	.25
Streambank—Agropyron riparium	95	85	.50
Tall—Agropyron elongatum	95	85	.50
Western—Agropyron smithii	80	80	.50
Wildrye:			
Canada—Elymus canadensis	90	60	1.00
Russian—Elymus junceus	90	70	1.00

* Indicates hard seeds may be present and are included.

3.2.1 *Noxious-weed seeds.* Agricultural seeds covered by this specification shall in no event contain more than 45 per pound of the following-named noxious-weed seed; bind-weed, Canada thistle, dodder, perennial sow-thistle, quackgrass, leafy spurge, whitetop, Johnsongrass, and Russian knapweed.

3.2.2 *Mixtures.* Mixtures of seeds shall be based upon the definite requirements for each kind of seed component, utilizing the name, percentage of pure seed, percentage of germination, and percentage of weed seeds given in table I. (See 6.1 and table II for example.)

3.3 Rejection. Seeds furnished under this specification which fail to meet the test requirements (allowing tolerances as provided under the Federal Seed Act) shall either be rejected or subjected to deductions in accordance with 6.2.

3.4 Workmanship. Agricultural seeds shall be thoroughly recleaned, and of uniformly good quality and appearance throughout each lot.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. Agricultural seeds shall be sampled by the procuring agency or an authorized representative at the time and place specified by the procuring agency in accordance with the regulations under the Federal Seed Act.

4.2 Inspection.

4.2.1 *General.* Inspection shall be made by the procuring agency or a duly authorized

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representative at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing, at any suitable point in transit, and/or after delivery to the point of destination.

4.2.2 *Responsibility for inspection.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.3 *Procedures.* The procedure shall be in accordance with the Regulations for Inspection and Certification of certain Agricultural Commodities and Products thereof (7CFR68) issued by the Secretary of Agriculture.

4.2.4 *Factory inspection.* The passing as satisfactory of any detail of processing or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.3 Tests. Purity and germination and noxious-weed seed tests shall be conducted in accordance with the methods prescribed in Service and Regulatory Announcements No. 156.

5. PREPARATION FOR DELIVERY

For civil agencies procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packing.

5.1.1 *Levels A and B.* Fifty or one-hundred pounds of seeds, as specified, shall be packed in textile waterproof laminated shipping bags conforming to PPP-B-35, type III. The bags shall be closed in conformance with the appendix to the bag specification (see 6.3).

5.1.2 *Level C.* The seed shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.2 Labeling and marking.

5.2.1 *Civil agencies.* Unless otherwise specified, the containers shall be labeled as required by the Federal Seed Act or applicable State seed law. The containers shall be marked in accordance with the requirements of Fed. Std. No. 123, and such other information as specified by individual agencies (see 6.3).

5.2.2 *Military agencies.* Unless otherwise specified, the containers shall be labeled as required by the Federal Seed Act or applicable State seed law. The containers shall be marked in accordance with MIL-STD-129 (see 6.3).

6. NOTES

6.1 The following example is given to show the method for determining requirements for mixtures from the requirements for each ingredient:

6.2 Deductions. Deductions in price may be based on the results of tests with tolerances being allowed, as compared to the percentage of pure seed, weed seed, and germination specified. The discount for straight kinds shall be as follows:

(1) Germination. Deduction of 2 percent for each percent of germination that the seed falls below the requirements after the allowance of the tolerance.

(2) Pure seed. Deduction of 2 percent for each 0.5 percent of pure seed that the seed

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TABLE II.—*Work table for determining requirements for mixtures from the requirements of each ingredient*

Weight in pounds of each ingredient	Percent by weight	Commodity	Percent pure seed specified	Minimum percent pure seed to be in mixture	Maximum percent weeds specified	Maximum percent weeds to be in mixture	Minimum percent germination specified
100	62-1/2	Ky. Bluegrass	85.00	53.13	1.00	0.625	80.00
50	31-1/4	Redtop	92.00	28.75	1.00	0.3125	90.00
10	6-1/4	White Clover	96.00	6.00	1.00	0.0625	90.00 (hard seed included)
Total 160				Total weed seed	1.00		

Purchasers should require the following from the above formula

	Pure seed	Minimum germination
	Percent	*Percent*
Ky. Bluegrass	53.13	80
Redtop	28.75	90
White Clover	6.00	90 (hard seed included)
Weeds not more than	1.00	

falls below the requirements after allowance of tolerance.

(3) Weed seeds. Deduction of 1 percent for each 0.20 percent or fraction thereof of weeds in excess of the amount specified after allowance of tolerances.

6.3 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Kind of seeds required (see 1.2.1).
(*c*) Level of packing required (see 5.1.1).
(*d*) Quantity of seeds desired (see 5.1.1).
(*e*) Other labeling and marking when required (see 5.2.1 and 5.2.2).

6.4 **Specification coverage.** The specification does not include all kinds and qualities of agricultural seeds that are commercially available but is intended to cover the kinds and quality generally suitable for Government requirements.

6.5 If samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.6 **Destination inspection.** When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.7 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Seeds, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Seeds, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY CUSTODIANS:

Army—MO

Navy—Docks

Air Force—MOA

Review Interest:

Army—MO

Navy Docks

Air Force—MOA

User Interest:

Army—GL

Copies of this specification may be purchased for 10 cents each.

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